MATH256
Linear Algebra HW#1

1. (1.2.2) Determine which matrices are in reduced echelon form and which others are only in echelon form.

a. $\begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

b. $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$

c. $\begin{bmatrix} 0 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$

d. $\begin{bmatrix} 0 & 1 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

2. (1.2.12) Find the general solutions of the systems whose augmented matrices are given

$$\begin{bmatrix} 1 & 0 & -9 & 0 & 4 \\ 0 & 1 & 3 & 0 & -1 \\ 0 & 0 & 0 & 1 & -7 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

3. (1.2.18) Determine the value(s) of h such that the matrix is the augmented matrix of a consistent linear system.

$$\begin{bmatrix} 1 & -3 & 1 \\ h & 6 & -2 \end{bmatrix}$$

4. (1.2.20) : Choose h and k such that the system has (a) no solution , (b) a unique solution , and (c) many solutions .Give the separate answers for each part

$$x_1 - 3x_2 = 1$$

$$2x_1 + hx_2 = k$$

5. (1.2.33) Find the interpolating polynomial p(t) = $a_0 + a_1 t + a_2 t^2$ for the data (1 , 6) , (2 , 15) ,(3 , 28) . That is , find a_0 , a_1 and a_2 such that

$$a_0 + a_1(1) + a_2(1)^2 = 6$$

$$a_0 + a_1(2) + a_2(2)^2 = 15$$

$$a_0 + a_1(3) + a_2(3)^2 = 28$$

6. (1.3.12) Determine if b is linear combination of a_1, a_2 and a_3

$$a_1 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \; a_2 = \begin{bmatrix} -2 \\ 3 \\ -2 \end{bmatrix}, \; a_3 = \begin{bmatrix} -6 \\ 7 \\ 5 \end{bmatrix}, \; b = \begin{bmatrix} 11 \\ -5 \\ 9 \end{bmatrix}$$

7. (1.3.16) Let $v_1 = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$, $v_2 = \begin{bmatrix} -2 \\ 1 \\ 7 \end{bmatrix}$ and y = $\begin{bmatrix} h \\ -3 \\ -5 \end{bmatrix}$. For what value(s) of h in the plane generated by $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$?

8. (1.3.18) List five vectors in Span$\{v_1, v_2\}$. For each vector, show the weights on $\boldsymbol{v_1}$ and $\boldsymbol{v_2}$ used to generate the vector and list the three entries of the vector. Do not make a sketch

$$v_1 = \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix}, \; v_2 = \begin{bmatrix} -2 \\ 3 \\ 0 \end{bmatrix}$$

9. (1.3.26) Let A = $\begin{bmatrix} 2 & 0 & 6 \\ -1 & 8 & 5 \\ 1 & -2 & 1 \end{bmatrix}$, let b = $\begin{bmatrix} 10 \\ 3 \\ 7 \end{bmatrix}$, and let W be the set of all linear combinations of the columns of A.

   a. Is **b** in W?
   b. Show that the second column of A is in W ?